



Unit 08: WRITING A PAPER

Scientific Research



I used to think about my sentences before writing them down; but ... I have found that it saves time to scribble in a vile hand whole pages as quickly as I possibly can ... Sentences thus scribbled down are often better ones than I could have written deliberately.

Charles Darwin
Autobiography



- A good principle is to **begin early**: if it is possible to start writing, then writing should start.
- Shaping the research and its outcomes into a write-up is an effective way of giving structure to a project, even if the work itself has not yet begun.
- The three main phases of a write-up are:
 - organizing materials so that the work tells a story,
 - giving this story the structure of a thesis or of an academic paper,
 - and actually writing.
- A paper or thesis, then, is an outcome of a cycle of activity, from speculation through definition and experimentation to write-up, with a range of obstacles and issues that can arise on the way.



I. THE SCOPE OF A PAPER

THE SCOPE OF A PAPER



- To begin a paper, the first task is to describe your aims.
- An effective exercise is to write down everything that motivated you to start the research.
 - What did you want to achieve?
 - What problems did you expect to address?
 - What makes the problems interesting?
- Next, define the scope of the work that you plan to write up.
 - To do so, it is necessary to make choices about what to include, and thus it is necessary to identify what *might* be included.

THE SCOPE OF A PAPER

- You may be reporting a particular piece of work, but the way it is reported is determined by the characteristics of the **audience**. The nature of the audience may even determine what can be reported.
- Making choices about the content of a paper places limits on its scope; these choices identify material to be excluded.
 - Broadly speaking, many research programs are a cycle of innovation and evaluation, with the answers or resolution of one investigation creating the questions that lead to the next.
 - But at some points it is necessary to stop undertaking new work and write up what has been achieved so far.
- Another element in the process of developing a paper is deciding **where the work might be published**.
 - In particular, the venue partly determines the scope of a paper.
 - For example, is there a page limit? Are there specific conventions to be observed? What prior knowledge or background is a reader likely to have? ...

TELLING A STORY



- A strong thesis or paper has a **story-like flow**, with a sequence of concepts building from a foundation of knowledge assumed to be common to all readers up to new ideas and results.
- The story told by a paper is a walk through the ideas and outcomes that explains the material in a structured way.
- A way to think about the starting point is to consider the “you” as you were the day that you began your research. Think about what you knew (and didn’t know) at that time, and what you have learnt since; your paper or thesis is a chance to teach the past “you” all the knowledge that is needed to become the current “you”.
- The early parts of the paper’s body typically explain hypotheses or claims; the reader expects to discover by the end whether these are justified.

STRUCTURING THE BODY OF A PAPER



There are several common ways for structuring the body of a paper, including:

- as a chain,
- by specificity,
- by example,
- and by complexity.

STRUCTURING THE BODY OF A PAPER

As a Chain

- Perhaps the most common structure is **a chain** in which the results and the background on which they build dictate a logical order for presentation of the material.
- First might come, say, a problem statement, then a review of previous solutions and their drawbacks, then the new solution, and finally a demonstration that the solution improves on its predecessors.

STRUCTURING THE BODY OF A PAPER

By Specificity

- One option is to structure **by specificity**, an approach that is particularly appropriate for results that can be divided into several stages.
- The material is first outlined in general terms, then the details are progressively filled in. Most technical papers have this organization at the high level, but it can also be used within sections.
- Material that might have such a structure is an explanation of a retrieval system. Such systems generally have several components. *For example*, in text retrieval a parser is required to extract words from the text that is being indexed; this information must be passed to a procedure for building an index; queries must likewise be parsed into a format that is consistent with that of the stored text; and a query evaluator uses the index to identify the records that match a given query. The explanation might begin with a review of this overall structure, then proceed to the detail of the elements.

STRUCTURING THE BODY OF A PAPER

By Example and By Complexity

- Another structure is **by example**, in which the idea or result is initially explained by, say, applying it to some typical problem. Then the idea can be explained more formally, in a framework the example has made concrete and familiar.
- A final alternative is to structure the body **by complexity**.
 - For example, a simple case can be given first, then a more complex case can be explained as an extension, thus avoiding the difficulty of explaining foundational concepts in a complex framework.
 - This approach is a kind of tutorial: the reader is brought by small steps to the full result.

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STRUCTURING THE BODY OF A PAPER



- When describing specific results, it is helpful, although not always possible, to begin with a brief overview of whatever has been observed.
- The rest of the discussion can then be used for amplification rather than further observations.
- News paper articles are often written in this “pyramid” style. The first sentence summarizes the story; the next few sentences review the story again, giving some context; then the remainder of the article presents the whole story in detail. Sections of research papers can sometimes be organized in this way.



II. ORGANIZATION



- Scientific papers follow a standard structure that **allows readers to quickly discover the main results**, and then, if interested, to examine the supporting evidence.
- A well-structured write-up accommodates this behavior by having important statements as near the beginning as possible.
- You need to:
 - Describe the work in the context of accepted scientific knowledge.
 - State the idea that is being investigated, often as a theory or hypothesis.
 - Explain what is new about the idea, what is being evaluated, or what contribution the paper is making.
 - Justify the theory, by methods such as proof or experiment.

ORGANIZATION



- Theses, journal articles, and conference papers have much the same organization when viewed in outline.
- A typical write-up has most of the following components:
 1. *Title and Author*
 2. *Abstract*
 3. *Introduction*
 4. *Literature Review*
 5. *Body*
 6. *Conclusions*
 7. *Bibliography*
 8. *Appendices*

1. TITLE AND AUTHOR

- Papers begin with their title and **information about authors** including name, affiliation, and address.
 - The convention in computer science is to not give your position, title, or qualifications;
 - but whether you give your name as A. B. Cee, Ae Cee, Ae B. Cee, or whatever, is a personal decision.
 - Use the same style for your name on all your papers, so that they are indexed together.
 - Include a durable email address or Web address.
- Another element is a collection of search terms, keywords, or key phrases - additional terminology that can be used to describe the topic of the paper.
 - Use words that concern the paper's principal themes.

2. ABSTRACT



- An abstract is typically a single paragraph of about 50–200 words.
- The function of an abstract is to allow readers to judge whether or not the paper is of relevance to them.
- It should therefore be a concise summary of the paper's aims, scope, and conclusions.
- Irrelevancies, such as minor details or a description of the structure of the paper, are usually inappropriate, as are acronyms, mathematics, abbreviations, or citations.

2. ABSTRACT

- Many abstracts follow a five-element organization:
 1. A general statement introducing the broad research area of the particular topic being investigated.
 2. An explanation of the specific problem (difficulty, obstacle, challenge) to be solved.
 3. A review of existing or standard solutions to this problem and their limitations.
 4. An outline of the proposed new solution.
 5. A summary of how the solution was evaluated and what the outcomes of the evaluation were.
- Thus a draft of an abstract can consist of five sentences, one for each of the points above.
- A valuable exercise is to read other papers, analyze their abstracts and introductions to see if they have this form, and then decide whether they are effective.

2. ABSTRACT

- The more specific an abstract is, the more interesting it is likely to be.
 - Instead of writing “space requirements can be significantly reduced”, for example, write “space requirements can be reduced by 60%”.
 - Instead of writing “we have a new inversion algorithm”, write “we have a new inversion algorithm, based on move-to-front lists”.
- Many scientists browse research papers outside their area of expertise. You should not assume that all likely readers will be specialists in the topic of the paper abstracts should be self-contained and written for as broad a readership as possible.

3. INTRODUCTION



- An introduction can be regarded as an **expanded version of the abstract**.
- It should describe the paper's topic, the problem being studied, references to key papers, the approach to the solution, the scope and limitations of the solution, and the outcomes.
- There needs to be enough detail to allow readers to decide whether or not they need to read further.
- It should include motivation: the introduction should explain why the problem is interesting, what the relevant scientific issues are, why the approach taken is a good one, and why the outcomes are significant.

3. INTRODUCTION

- The introduction can discuss the importance or ramifications of the conclusions but should include only a brief summary of the supporting evidence, which the interested reader can find in the body of the paper.
- The introduction should clearly tell the reader what in the paper is new and what the outcomes are.
 - If, however, the existence of results is concealed until later on, the reader might assume there are no results and discard the paper as worthless.
- The reader should also understand the contribution, that is, what the *discovery* of the work is—the core idea that the referees or examiners need to appreciate as novel and important.

- The body of a paper should present the results.
- This presentation should:
 - provide necessary background and terminology,
 - explain the chain of reasoning that leads to the conclusions,
 - provide the details of central proofs,
 - summarize any experimental outcomes,
 - and state in detail the conclusions outlined in the introduction.

4. BODY

- **Descriptions of experiments** should permit reproduction and verification.
- There should be **careful definitions of the hypothesis** and major concepts, even those that were described informally in the introduction.
- The structure should be evident in the section headings. Since the body can be long, narrative flow and a clear logical **structure are essential**.
- The body should be reasonably **independent of other papers**.
 - If, to understand your paper, the reader must find specialized literature such as your earlier papers or an obscure paper by your advisor, then its audience will be limited.

- Consider a detailed example: an investigation of external sorting in database systems. In this task, a large database table—tens of millions of records, say, constituting many gigabytes—must be sorted on a field that is specified in a query. An effective sorting method is to sort the table one block at a time, storing the sorted blocks in a temporary file then merging them to give the final result. Costs include processing time for sorting and merging, transfer time to and from disk, and temporary space requirements. The balance between these costs is governed by available in memory buffer space, as large blocks are expensive to sort but cheap to merge. The specific research question being investigated is whether disk costs can be reduced by compressing the data while it is sorted.

4. BODY

- Components of the body might include, among other things, background, previous work, proposals, experimental design, analysis, results, and discussion.
- Specific research projects suggest specific headings. For the “compression for fast external sorting” project, the complete set of section headings might be:
 1. Introduction
 2. External sorting
 3. Compression techniques for database systems
 4. Sorting with compression
 5. Experimental setup
 6. Results and discussion
 7. Conclusions
- The wording of these headings does not follow the standard form, but the intent of the wording is the same. Sections 2 and 3 are the background; Section 4 contains novel algorithms, and Sections 4 and 5 together are the methods.

4. BODY

- The **background material** can be entirely separate from the discussion of **previous work** on the same problem.
- The former is the knowledge the reader needs to understand your contribution.
- The latter is, often, alternative approaches that are superseded by your work.
- Together, the discussion of background and previous work also introduce the state of the art and its failings, the importance and circumstances of the research question, and benchmarks or baselines that the new work should be compared to.

4. BODY

- The **results section** is an assembly of evidence on which the key arguments are based.
- This typically includes presentations of experimental outcomes, theorems, proofs, analyses of data, and tabulations of investigative outcomes and discoveries.
- The arguments then convey what the results mean—that is, they need to be explained, analyzed, and interpreted.
- For the data, readers need to know how the data was gathered, how they might obtain or create the data for themselves, and background on issues such as limitations of the data.

5. LITERATURE REVIEW

- Few results or experiments are entirely new. Usually they are extensions of or corrections to previous research—that is, most results are an incremental addition to existing knowledge.
- A literature review, or survey, is used to compare the new results to similar previously published results, to describe existing knowledge, and to explain how it is extended by the new results.
- A survey can also help a reader who is not expert in the field to understand the paper and may point to standard references such as texts or survey articles.

5. LITERATURE REVIEW

- The literature review can be **early in a paper**, to describe the context of the work, and might in that case be part of the introduction; or, less commonly, the literature review can follow or **be part of the main body**, at which point a detailed comparison between the old and the new can be made. If the literature review is **late in a paper**, it is easier to present the surveyed results in a consistent terminology, even when the cited papers have differing nomenclature and notation.
- In some papers the literature review material is not gathered into a single section, but is discussed where it is used—background material in the introduction, analysis of other researchers' work as new results are introduced, and so on. This approach can help you to write the paper as a flowing narrative, but makes it difficult for a reader to assess your depth of understanding of the field.

6. CONCLUSIONS

- The conclusions section, or summary, is used to draw together the topics discussed in the paper.
- This section should include a concise statement of the paper's **important results** and an **explanation of their significance**.
- This is an appropriate place to state (or restate) any **limitations of the work**: shortcomings in the experiments, problems that the theory does not address, and so on.
- The conclusions are an appropriate place to **look beyond the current context** to other problems that were not addressed, to questions that were not answered, to variations that could be explored.

BIBLIOGRAPHY AND APPENDICES



7. *Bibliography*

- A paper's bibliography, or its set of references, is a complete list of theses, papers, books, and reports cited in the text. No other items should be included.

8. *Appendices*

- Some papers have appendices giving detail of proofs or experimental results, and, where appropriate, material such as listings of computer programs.
- The purpose of an appendix is to hold bulky material that would otherwise interfere with the narrative flow of the paper, or material that even interested readers do not need to refer to.
- Appendices are only occasionally necessary for a paper, in cases where there is material such as a proof whose length would interrupt the flow.



III. FROM DRAFT TO SUBMISSION

THE FIRST DRAFT



- For the first draft of a write-up you may find it helpful to **write freely**—without particular regard to style, layout, or even punctuation—so that you can concentrate on presenting a smooth flow of ideas in a logical structure.
- Worrying about how to phrase each sentence tends to result in text that is clear but doesn't form a continuous whole, and authors who are too critical on the first draft are often unable to write anything at all.
- A consequence of having a sloppy first draft is that you must **edit and revise carefully**; initial drafts are often awkwardly written and full of mistakes. But few authors write well on the first draft anyway. The best writing is the result of frequent, thorough revision.

THE FIRST DRAFT



- Mathematical content, definitions, and **the problem statement** should be made precise as early as possible.
- The hypothesis and the results flow from a clear statement of the problem being tackled.
- Describing the problem forces you to consider in depth the scope and nature of the research.
- If you find that you cannot describe the problem precisely, then perhaps your understanding is lacking, or the ideas are insufficiently developed.

THE FIRST DRAFT



- The writing defines the research, and the one cannot proceed without the other.
- **Writing is a stimulus to research**, suggesting fresh ideas and clarifying vague concepts and misunderstandings; and development of the presentation of the results often suggest the form the proofs or experiments should take.
- Research is also a stimulus to writing—fine points are quickly forgotten once the work is complete.
- Don't expect the writing to progress steadily but do expect progress overall. If the writing seems to have stalled, it is time to put other tasks aside for a while.

FROM DRAFT TO SUBMISSION



- One of the techniques used for composing a technical paper is to brainstorm, writing down in point form what has been learnt, what has been achieved, and what the results are.
- The next step is to prepare a skeleton, choosing results to emphasize and discarding material that on reflection seems irrelevant, and then work out a logical sequence of sections that leads the reader naturally to the results.

FROM DRAFT TO SUBMISSION



- A useful discipline is to choose the section titles before writing any text, because if material to be included doesn't seem to belong in any section then the paper's structure is probably faulty.
- The introduction is completed first and includes an overview of the paper's intended structure, that is, an outline of the order and content of the sections.
- When the body and the closing summary are complete, the introduction usually needs substantial revision because the arguments presented in the paper are likely to mature and evolve as the writing proceeds.
- The final version of the abstract is the last part to be written.

FROM DRAFT TO SUBMISSION



- **With a reasonably thorough draft completed**, it is time to review the write-up's content and contribution.
- Anticipate likely concerns or objections that the referees may have, and address them; if they can't be addressed, acknowledge them.
- Consider whether extra work is needed to fill a hole.
- Ask the probing, critical questions that you would ask of other people's work.
- The burden of proof is on you, not the reader, so be conservative in your claims and thorough with your evidence.

FROM DRAFT TO SUBMISSION



- During drafting and revision, **ensure that the topic of the paper does not drift.**
 - At the start of the writing process, you wrote down your aims, motivation, and scope. Use these as a reference.
 - If you feel that you need to write something that is not obviously relevant to your original aims, then either establish the connection clearly or alter the aims.
 - Changing the aims can affect the work in many ways, however, so only do so with great care.

REFERENCES



- Zobel, J. (2014). *Writing for computer science*. London: Springer.